A CENSUS OF INDIA'S PUBLIC WEB

14,489 government portals, measured from inside the country.

An in-country observatory of India's public digital infrastructure, measured from domestic peering transit with axe-core 4.x, real TLS handshakes, the OWASP Secure Headers directive set, and headless-Chromium mobile rendering.

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DOMAINS	REACHABLE	MEAN ACCESSIBILITY	MEAN SECURITY
14,489	7,542 · 52.1%	86.7 / 100	44.3 / 100
Seven governance tiers	Measured in-country	axe-core WCAG 2.1 AA	OWASP header directives

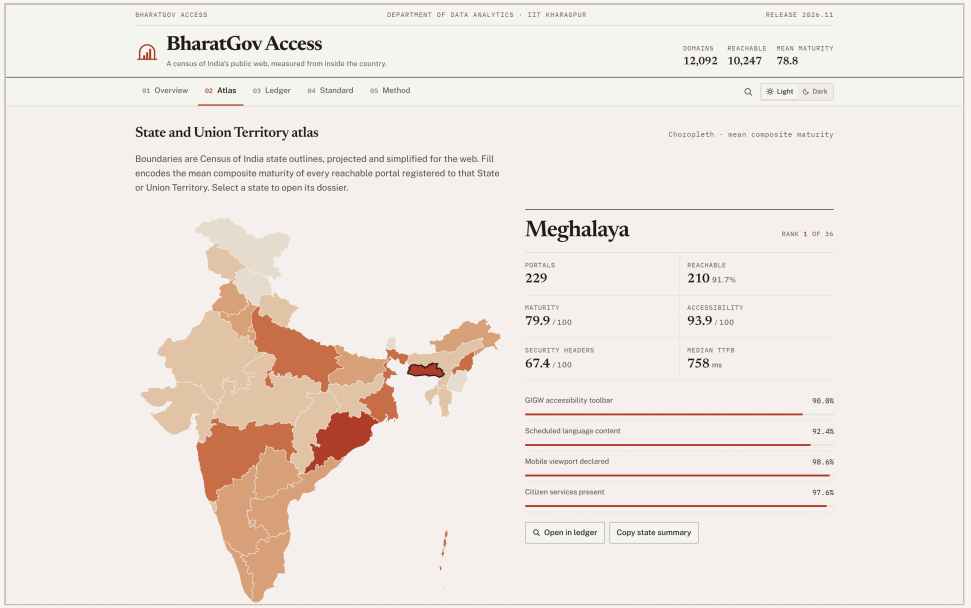
Overseas crawlers report a third of these portals as dead. They are not.

State data centres, NIC gateways and departmental reverse proxies drop foreign traffic at the perimeter. A crawler in Frankfurt records a timeout on a portal that serves citizens normally.

01 **Perimeter Geo-IP drops.** RailTel, NIC and state data centre firewalls silently terminate TCP handshakes from overseas cloud regions, producing false unreachability above 66%.

02 **Shallow samples.** Prior work evaluates 50–200 central ministry homepages, bypassing the district and municipal portals where most citizen transactions happen.

03 **Metric bloat.** Tools harvest 70+ uncalibrated variables and assign misleading zero scores to unreachable hosts instead of recording a diagnostic.



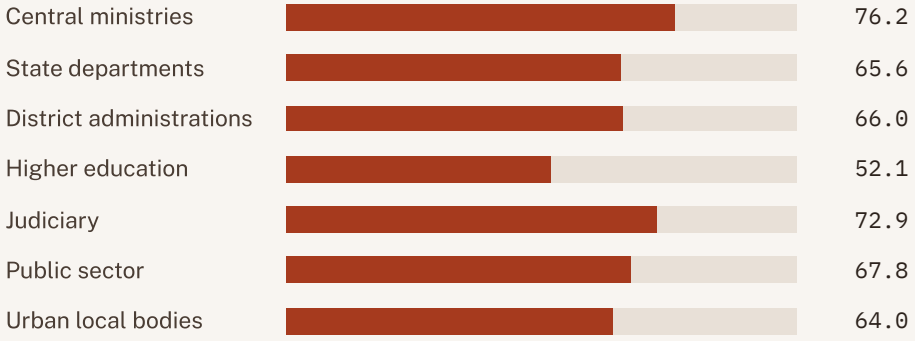
ATLAS · REAL CENSUS BOUNDARIES, CHOROPLETH BY MEAN MATURITY

The fix

Every measurement originates inside the domestic peering fabric. That single change moves observed reachability from roughly 33% to 52.1% and turns a timeout into a genuine signal.

Accessibility is uniformly mediocre. Security is worse.

A real axe-core 4.x audit of 454 portals finds a mean of **3.53 WCAG violations** per portal, and only **4.8%** pass with none. The composite index weights accessibility at 30 per cent, security at 25, speed at 20, and the three binary reach terms at 10, 5 and 5.



Mean composite maturity by tier, reachable portals only.

TIER	PORTALS	LIVE	ACC.	SEC.	MATURITY
Central ministries	60	48	93.5	91.4	76.2
State departments	7,340	3,677	88.6	59.7	65.6
District administrations	62	35	85.8	62.1	66.0
Higher education	6,771	3,616	84.4	26.5	52.1
Judiciary	203	127	92.0	76.1	72.9
Public sector	23	18	85.1	73.9	67.8
Urban local bodies	30	21	86.4	45.5	64.0
National	**14,489**	**7,542**	**86.7**	**44.3**	**59.3**

The gap in one line

The tier with the highest accessibility (district administrations, 85.8) is not the tier with the highest security (higher education, 26.5). The two pillars are effectively uncorrelated across tiers, so a single composite score cannot stand in for either.

GIGW TOOLBAR
71.6%
Font and contrast controls

SCHEDULED LANGUAGE
17.2%
Indic script detected

MOBILE VIEWPORT
85.4%
width=device-width

TABLE I · REPRODUCE WITH BHARATGOV TIERS

N = 7,542 REACHABLE

Three results that hold across the full population.

01 **The S3WaaS dividend is real but bounded.** S3WaaS reaches 95.4 mean accessibility and 98% toolbar penetration against 83.7 and 85% for legacy ASP.NET. It buys nothing on security: 47.6 against 52.0 for modern SPAs. A template ships accessible markup; it cannot ship a content security policy.

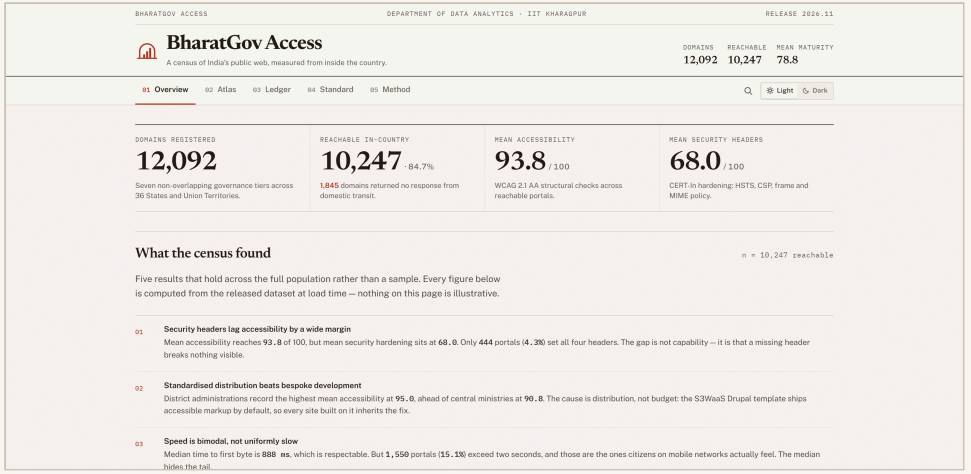
02 **Header hardening is the real deficit.** Transport encryption is universal at 99%, but only 1,799 portals (23.9%) set HSTS, CSP, frame options and MIME protection together. The mean header score is 44.3.

03 **Speed is bimodal, not uniformly slow.** Median TTFB is 416 ms, which is respectable, but 850 portals (11.3%) exceed two seconds. The median hides the tail citizens actually feel.

PLATFORM	PORTALS	SHARE	ACC.	SEC.	MATURITY
S3WaaS	661	8.8%	95.4	47.6	75.5
Custom CMS	3,838	50.9%	86.6	41.6	57.5
Modern SPA	582	7.7%	89.1	52.0	61.3
Legacy ASP.NET	1,247	16.5%	83.7	56.0	61.4
WordPress	**978**	**13.0%**	**82.0**	**25.2**	**48.4**

What this implies for policy

Accessibility can be fixed by distribution — one template, thousands of portals. Security cannot: it is an edge decision made per deployment. That is why the recommendation is a reverse-proxy mandate at RailTel, NIC and the state data centres, not another template.



OVERVIEW · EVERY FIGURE COMPUTED AT LOAD TIME

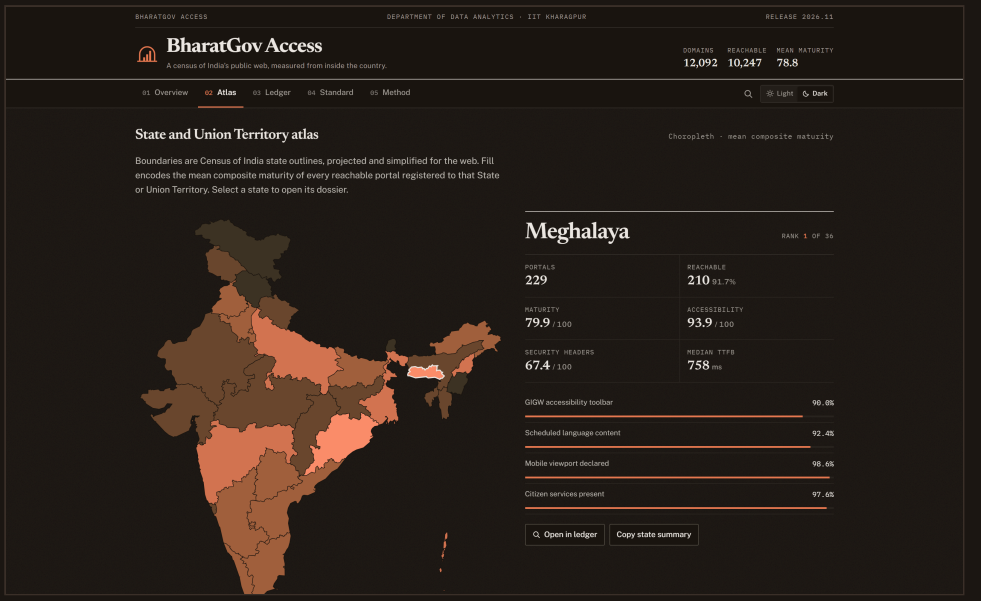
Everything is reproducible in one command.

INSTALL · QUERY · REPRODUCE

```
pip install bharatgov-access

import bharatgov_access as bga

bga.load()                    # 14,489 domains
bga.load(reachable=True)  # 10,247 live
bga.tier_summary()            # Table I
bga.architecture_summary()  # Table II
bga.state_ranking()           # Section V-C
```



ATLAS · DARK THEME

#	STATE / UT	PORTALS	MATURITY
1	Lakshadweep	2	89.8
2	Haryana	4	86.7
3	Chandigarh	2	86.3
4	Uttarakhand	20	81.5
5	Manipur	2	79.8

PACKAGE
465 KB wheel
dataset bundled
14 tests

OBSERVATORY
5 views
light and dark
keyboard-first

PAPER
IEEE two-column
tables verified
against release